Mathematical Studies Analysis

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Fall 2026

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1 The fundamental theorem of calculus

1.1 Differentiation of monotone functions

Continuity

Definition. Let X be a set and $g : X \to [0, \infty)$. Define infinite sum

$$\sum_{x \in X} g(x) \coloneqq \left\{ \sum_{x \in Y} g(x) : Y \subset X \text{ finite} \right\}.$$

Theorem. Let X be a set and $g : X \to [0, \infty)$. If $\sum_{x \in X} g(x) < \infty$, then $\{x \in X : g(x) > 0\}$ is countable.

Proof. Let $M = \sum_{x \in X} g(x)$ and $X_k = \{x \in X : g(x) > k^{-1}\}$ for $k \in \mathbb{N}$. Let $Y \subset X_k$ finite, then

$$M \geq \sum_{x \in Y} g(x) > k^{-1} |Y| .$$

Hence $|Y| \leq kM$ and thus $|X_k| \leq kM$. It follows that $\{x \in X : g(x) > 0\} = \bigcup_{k=1}^{\infty} X_k$ is countable. □

Theorem. Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ an increasing function. Then, f is continuous at all but countably many $x \in I$.

Proof. First assume $I = [a, b]$. Then for $x \in (a, b)$, there exists $\lim_{y \to x^-} f(y) = f(x-)$ and $\lim_{y \to x^+} f(y) = f(x+)$. Since the function is increasing, $f(x+) \geq f(x-)$. Now we can define $g(x) = f(x+) - f(x-)$. It is clear that $\sum_{x \in [a,b]} g(x) \leq f(b) - f(a)$. By the previous theorem, $f(x+) = f(x-)$ for all but countably many $x \in [a, b]$.

The interval $I \subset \mathbb{R}$ can be written as a countable union of finite closed intervals. The proof is then complete. □

Differentiation

Theorem (Lebesgue's differentiation theorem)**.** Let $I \subset \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be a monotone function. Then f is differentiable a.e. on I.

To show this, we need a few lemmas.

Lemma. Let $I \subset \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let

$$E = \left\{ x \in I^\circ : \text{ there exist } f'_+(x) \text{ and } f'_-(x) \text{ and } f'_+(x) \neq f'_-(x) \right\}.$$

Then E is countable.

Proof. First observe that for any function $g : I \to \mathbb{R}$, the set of its strict local minimum points in I° is countable. Set

$$E_- = \left\{x \in E : f'_-(x) < f'_+(x)\right\}.$$

For every $r \in \mathbb{Q}$, define

$$E_r = \left\{x \in E_- : f'_-(x) < r < f'_+(x)\right\},$$

and let $g_r(t) = f(t) - rt$ for $t \in I$. By density of rationals, $E_- = \bigcup_{r\in\mathbb{Q}} E_r$. If $x \in E_r$, then

$$(g_r)'_-(x) < 0 < (g_r)'_+(x).$$

It follows that $g_r(y) > g_r(x)$ for every $y \neq x$ sufficiently close to x, i.e. x is a strict local minimum point of g_r. Therefore E_r is countable and thus E_- is countable. Applying the same argument to $-f$ completes the proof for the lemma. □

Lemma. Let $E \subset \mathbb{R}$ be bounded, and $\mathcal{F}$ be a family of open intervals such that each $x \in E$ is the left endpoint of an interval $(x, x + h_x) \in \mathcal{F}$. Then there for every $\varepsilon > 0$, there exist disjoint intervals $I_1, \ldots, I_n \in \mathcal{F}$ such that

$$\mathcal{L}^1_o\left(E \cap \bigcup_{k=1}^{n} I_k\right) \geq \mathcal{L}^1_o(E) - \varepsilon.$$

Proof. If $E = \emptyset$, take the empty collection. If $E \neq \emptyset$ and $\mathcal{L}^1_o(E) \leq \varepsilon$, any one interval in $\mathcal{F}$ suffices. Thus assume that $L^1_o(E) > \varepsilon$. For each $x \in E$, choose an interval $(x, x + h_x) \in \mathcal{F}$ and define

$$E_n = \left\{x \in E : h_x > n^{-1}\right\}.$$

Note that $E_n \subset E_{n+1}$ and $E = \bigcup_{n=1}^{\infty} E_n$. By coninuity from below of Lebesgue outer measure (exercise), we have $\mathcal{L}^1_o(E_n) \to \mathcal{L}^1_o(E)$. Hence we can find m large enough such that $\mathcal{L}^1_o(E_m) > \mathcal{L}^1_o(E) - \frac{\varepsilon}{2}$. For each $k \in \mathbb{N}$, let $\eta_k = \varepsilon 2^{-k-1}$, then $\sum_{k=1}^{\infty} \eta_k = \frac{\varepsilon}{2}$.

Let $a_1 = \inf E_m$. By definition of infimum we can find $x_1 \in E_m$ with $a_1 \leq x_1 < a_1 + \eta_1$, and correspondingly $I_1 = (x_1 + h_1) \in \mathcal{F}$ with $h_1 > m^{-1}$. Suppose $I_1, \ldots, I_k$ has been chosen. If $\{x \in E_m : x \geq x_k + h_k\} = \emptyset$, we stop. Otherwise let $a_{k+1} = \inf\{x \in E_m : x \geq x_k + h_k\}$. Then there is $x_{k+1} \in E_m$ such that $a_{k+1} \leq x_{k+1} \leq a_{k+1} + \eta_{k+1}$ and correspondingly $I_{k+1} = (x_{k+1}, x_{k+1} + h_{k+1}) \in \mathcal{F}$ with $h_{k+1} > m^{-1}$. Note that

$$x_{k+1} \geq x_k + h_k > x_k + m^{-1},$$

and $E_m \subset E$ is bounded, the construction must stop after finitely many steps. Moreover, $I_1, \ldots, I_n$ are disjoint by construction.

Now let

$$S = \bigcup_{k=1}^{n} I_n, \quad T = \bigcup_{k=1}^{n} (x_k - \eta_k, x_k].$$

The points of E_m skipped at the k-th step belong to $[a_k, x_k] \subset (x_k - \eta_k, x_k]$. Thus $E_m \subset S \cup T$, then

$$\begin{aligned}\mathcal{L}^1_o(E) - \frac{\varepsilon}{2} < \mathcal{L}^1_o(E_m) &\leq \mathcal{L}^1_o(E_m \cap S) + \mathcal{L}^1_o(E_m \cap T) \\ &\leq \mathcal{L}^1_o(E_m \cap S) + \sum_{k=1}^{n} \eta_k \leq \mathcal{L}^1_o(E_m \cap S) + \frac{\varepsilon}{2}.\end{aligned}$$

This concludes the proof. □

Lemma. Let $E \subset \mathbb{R}$ be bounded, $\mathcal{F}$ be family of open intervals such that for all $x \in E$ and $\delta > 0$, there exists $(x, x + h_x) \in \mathcal{F}$ with $0 \leq h_x < \delta$. Then for all $\varepsilon > 0$, there exists disjoint intervals $I_1, \ldots, I_m \in \mathcal{F}$ such that

$$\mathcal{L}^1_o\left(E \cap \bigcup_{i=1}^{m} I_i\right) \geq \mathcal{L}^1_o(E) - \varepsilon, \quad \sum_{i=1}^{m} \mathcal{L}^1(I_i) \leq \mathcal{L}^1_o(E) + \varepsilon.$$

Proof. By the definiton of Lebesgue outer measure, find an bounded open set $U \supset E$ such that $\mathcal{L}^1_o(U) \leq \mathcal{L}^1_o(E) + \varepsilon$. Let $\mathcal{F}'$ be the subfamily of intervals $(x, x + h_x) \in \mathcal{F}$ contained in U. Note that for each $x \in E \subset U$, there must exists at least one such interval, since U contains a ball centered at x and there are intervals of arbitrarily small length.

Apply the previous lemma to the family $\mathcal{F}'$ to find disjoint intervals $I_1, \ldots, I_n \subset \mathcal{F}'$ such that

$$\mathcal{L}^1_o\left(E \cap \bigcup_{k=1}^{n} I_k\right) \geq \mathcal{L}^1_o(E) - \varepsilon.$$

Since the intervals are disjoint and contined in U, it follows that

$$\sum_{k=1}^{n} \mathcal{L}^1(I_k) \leq \mathcal{L}^1_o(U) \leq \mathcal{L}^1_o(E) + \varepsilon.$$

The proof is then complete. □

Definition. For a function $f : E \to \mathbb{R}$, define

$$\liminf_{x \to x_0} f(x) = \sup_{r>0} \inf_{0<d(x,x_0)<r} f(x), \quad \limsup_{x \to x_0} f(x) = \inf_{r>0} \sup_{0<d(x,x_0)<r} f(x),$$

Exercise. $\liminf_{x \to x_0} f(x) = l \in \mathbb{R}$ iff for every $\varepsilon > 0$:

1. there exists $\delta > 0$ such that $l - \varepsilon < f(x)$ for all $x \in E$ with $0 < d(x, x_0) < \delta$.
2. For every $\delta > 0$, there exists $x \in E$ with $0 < d(x, x_0) < \delta$ such that $f(x) < l + \varepsilon$.

Also, $\liminf_{x \to x_0} f(x) = -\infty$ iff for every $M > 0$ and $\delta > 0$, there exists $x \in E$ with $0 < d(x, x_0) < \delta$ such that $f(x) < -M$.

Now we are ready to show Lebesgue's differentiation theorem. Recall the statement of the theorem below:

Theorem (Lebesgue's differentiation theorem)**.** Let $I \subset \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be a monotone function. Then f is differentiable a.e. on I.

Proof. Define **Dini's derivatives** as follows:

$$\overline{D}_+ f(x_0) = \limsup_{x \to x_0^+} \frac{f(x) - f(x_0)}{x - x_0}, \qquad \underline{D}_+ f(x_0) = \liminf_{x \to x_0^+} \frac{f(x) - f(x_0)}{x - x_0},$$
$$\overline{D}_- f(x_0) = \limsup_{x \to x_0^-} \frac{f(x) - f(x_0)}{x - x_0}, \qquad \underline{D}_- f(x_0) = \liminf_{x \to x_0^-} \frac{f(x) - f(x_0)}{x - x_0}.$$

WLOG assume f is increasing, and assume $I = [a, b]$. Let

$$\begin{aligned} E &= \left\{x \in I^\circ : \underline{D}_+ f(x) < \overline{D}_+ f(x)\right\} \\ &= \bigcup_{\substack{r,s \in \mathbb{Q} \\ 0<r<s}} \left\{x \in I^\circ : \underline{D}_+ f(x) < r < s < \overline{D}_+ f(x)\right\} \\ &= \bigcup_{\substack{r,s \in \mathbb{Q} \\ 0<r<s}} E_{r,s} \end{aligned}$$

It suffices to show that $\mathcal{L}^1(E_{r,s}) = 0$. Let $\varepsilon > 0$ be given.

Let $x \in E_{r,s}$. We have

$$\liminf_{h \to 0^+} \frac{f(x+h) - f(x)}{h} < r.$$

Then for each $\delta > 0$, there exists $0 < h_x < \delta$ such that $h_x^{-1}(f(x + h_x) - f(x)) < r$. Let $\mathcal{F}$ be the collection of $(x, x + h_x)$, where h_x is selected as aforementioned while δ ranges over all positive value, and x ranges over $E_{r,s}$. It is then clear that $\mathcal{F}$ satisfies the condition for the previous lemma.

Apply previous lemma, there exists $I_1, \ldots, I_m \in \mathcal{F}$ disjoint such that

$$\mathcal{L}^1_o\left(E_{r,s} \cap \bigcup_{i=1}^m I_i\right) \geq \mathcal{L}^1_o(E_{r,s}) - \varepsilon, \tag{1}$$

$$\sum_{i=1}^m \mathcal{L}^1(I_i) \leq \mathcal{L}^1_o(E_{r,s}) + \varepsilon. \tag{2}$$

Write $I_i = (x_i, x_i + h_i)$, then

$$\sum_{i=1}^n f(x_i + h_i) - f(x_i) < r \sum_{i=1}^n h_i \leq r\mathcal{L}^1_o(E_{r,s}) + r\varepsilon. \tag{3}$$

Now let $V = \bigcup_{i=1}^{m}(x_i, x_i + h_i)$, which is open. If $x \in E_{r,s} \cap V$, we have

$$s < \limsup_{t \to 0} \frac{f(x+t) - f(x)}{t}.$$

Then for every $\delta > 0$, there exists f_x such that $0 < t_x < \delta$ and $s < t_x^{-1}(f(x + t_x) - f(x))$. Define $\mathcal{G}$ to be the collection of $(x, x + t_x)$ that is contained in V, where t_x is selected as aforementioned while $\delta > 0$ ranges over all positive value, and x ranges over $E_{r,s} \cap V$.

The previous lemma again gives $J_1, \ldots, J_l \in \mathcal{G}$ disjoint such that

$$\mathcal{L}_o^1\left(E_{r,s} \cap V \cap \bigcup_{k=1}^{l} J_k\right) \geq \mathcal{L}_o^1(E_{r,s} \cap V) - \varepsilon.$$

Write $J_k = (y_k, y_k + t_k)$, we have

$$\begin{aligned}\sum_{k=1}^{l} f(y_k + t_k) - f(y_k) &\geq s \sum_{k=1}^{l} t_k \\ &\geq s\mathcal{L}_o^1(E_{r,s} \cap V) - s\varepsilon \\ &\geq s\mathcal{L}_o^1(E_{r,s}) - 2s\varepsilon. \end{aligned} \tag{4}$$

Now compare (3) and (4). Since f is increasing and each J_k lies in one of the interval I_i, it is not hard to see that

$$\sum_{i=1}^{n} f(x_i + h_i) \geq \sum_{k=1}^{l} f(y_k + t_k) - f(y_k).$$

It follows that

$$\mathcal{L}_o^1(E_{r,s}) \leq \frac{2s + r}{s - r}\varepsilon.$$

Letting $\varepsilon \to 0$ completes the proof for $\mathcal{L}^1(E_{r,s}) = 0$. This implies that $f'_+(x)$ exists for a.e. $x \in [a, b]$. Similarly, $f'_-(x)$ exists for a.e. $x \in [a, b]$. A previous lemma shows that the left and right derivatives are equal a.e. Therefore, $f'(x)$ exists a.e.

It remains to show the left and right derivatives are a.e. finite. Define

$$g(x) = \begin{cases} f(x), & x \in [a, b] \\ f(b), & x \geq b. \end{cases}$$

Define also

$$g_n(x) = \frac{g(x + n^{-1}) - g(x)}{n^{-1}}$$

for $x \in [a,b]$ and $n \in \mathbb{N}$. Then, $g_m(x) \to f'(x)$ for a.e. $x \in [a,b]$. Since f is monotone, it is measurable and thus g_m is also measurable. By Fatou's lemma,

$$\begin{aligned}
\int_a^b f'(x)\,dx &\le \liminf_{n\to\infty} \int_a^b g_n(x)\,dx \\
&= \liminf_{n\to\infty} n\left(\int_a^b g(x+n^{-1})\,dx - \int_a^b g(x)\,dx\right) \\
&= \liminf_{n\to\infty} n\left(\int_a^{b-n^{-1}} g(x+n^{-1})\,dx + \int_{b-n^{-1}}^b f(b) - \int_a^b g(x)\,dx\right) \\
&= \liminf_{n\to\infty} n\left(\int_{a+n^{-1}}^b g(t)\,dt - \int_a^b g(t)\,dt\right) + f(b) \\
&= \liminf_{n\to\infty} f(b) - n\int_a^{a+n^{-1}} f(t)\,dt \\
&\le f(b) - f(a).
\end{aligned}$$

Therefore, f' is integrable, which implies $\{f' = \infty\}$ has measure zero.

For a general interval $I \subset \mathbb{R}$, I can be written as a countable union of bounded interval, the proof for the theorem is complete. □

Remark. From the proof above, we see that for monotone function $f : [a,b] \to \mathbb{R}$, we have

$$\int_a^b f' \le f(b) - f(a).$$

There exists a monotone function for which inequality is strict. Indeed, let $f : [0,1] \to [0,1]$ be the Cantor function. Then, $f' = 0$ a.e. but $f(1) = 1$ and $f(0) = 0$.

1.2 Differentiation of BV functions

Definition (pointwise variation)**.** Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}^N$. The **pointwise variation** of f on the interval I is defined as

$$\operatorname{Var} f = \sup\left\{\sum_{i=1}^n \|f(x_i) - f(x_{i-1})\|\right\},$$

where the supremum is taken over all partition $x_0 < x_1 < \cdots < x_n$, where $n \in \mathbb{N}$ and $x_0, \ldots, x_n \in I$. Say a function has **bounded pointwise variation** if $\operatorname{Var} f < \infty$.

The space of all functions $f : I \to \mathbb{R}^N$ with bounded pointwise variation is denoted by $BV(I;\mathbb{R}^N)$.

Proposition. The following properties holds:

1. Fix $c \in I$, then
$$\|f\|_{BV} = \operatorname{Var} f + \|f(c)\|$$
is a norm on $BV(I;\mathbb{R}^N)$. Moreover, $BV(I;\mathbb{R}^N)$ is complete under this norm.
2. Let $c \in I$, then
$$\operatorname*{Var}_I f = \operatorname*{Var}_{I\cap(-\infty,c]} f + \operatorname*{Var}_{I\cap[c,\infty)} f.$$

Proof. Exercise. □

Definition. Let $f : I \to \mathbb{R}^N$. Say f has **locally bounded variation** if $\operatorname{Var}_{[a,b]} f < \infty$ for all $[a,b] \subset I$. Denote the space of functions with bounded local variation as $BV_{\text{loc}}(I;\mathbb{R}^N)$.

Example. Let $f : I \to \mathbb{R}^N$ is Lipshitz continuous with Lipshitz constant L, $[a,b] \subset I$, and $a \le x_0 < x_1 < \cdots < x_n < b$. Then,
$$\sum_{i=1}^{n} \|f(x_i) - f(x_{i-1})\| \le L \sum_{i=1}^{n} (x_i - x_{i-1}) \le L(b-a).$$
Therefore, Lipshitz functions has locally bounded variation.

Exercise. Let $f, g \in BV([a,b])$, then the following holds:

1. $fg \in BV([a,b])$,
2. Further suppose $g \neq 0$, then fg^{-1} needs not be in $BV([a,b])$. However, if there exists $c > 0$ such that $g(x) \ge c$ for all $x \in I$, we have $fg^{-1} \in BV([a,b])$.

Theorem. Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ increasing. Then,
$$\operatorname{Var} f = \sup f - \inf f.$$
In particular, $f \in BV_{\text{loc}}(I)$, and if f is bounded, $f \in BV(I)$.

Proof. Let $x_0 < x_1 < \cdots < x_n$ be a partition of I, then
$$\sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| = f(x_n) - f(x_0) \le \sup f - \inf f.$$
Hence, $\operatorname{Var} f \le \sup f - \inf f$. Conversely, assume $\inf I < \sup I$ and take $\inf I < a < b < \sup I$. Then, we have $f(b) - f(a) \le \operatorname{Var} f$. Letting $a \to \inf I$ and $b \to \sup I$ completes the proof. □

Theorem. Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}^N$. Fix $c \in I$ and define
$$v(x) = \begin{cases} \operatorname{Var}_{[c,x]} f & (x \ge c), \\ -\operatorname{Var}_{[x,c]} f & (x < c). \end{cases}$$
Then, for every $x, y \in I$ with $x < y$, $\|f(x) - f(y)\| \le v(y) - v(x)$.

Moreover, if $f \in BV_{\text{loc}}(I; \mathbb{R}^N)$, then the left and right limit exists for all $x \in I^\circ$, and f is continuous at all but countably many points. Also, if $N = 1$, $v \pm f$ are increasing.

Proof. Let $x < y$, then

$$\|f(y) - f(x)\| \le \operatorname*{Var}_{[x,y]} f = \begin{cases} \operatorname{Var}_{[c,y]} f - \operatorname{Var}_{[c,x]} f & (c \le x < y), \\ \operatorname{Var}_{[x,c]} f - \operatorname{Var}_{[y,c]} f & (x < y \le c), \\ \operatorname{Var}_{[x,c]} f + \operatorname{Var}_{[c,y]} f & (x \le c \le y). \end{cases}$$

It directly follows that $\|f(x) - f(y)\| \le v(y) - v(x)$.

Now assume $f \in BV_{\text{loc}}(I; \mathbb{R}^N)$, then v is real-valued. Now that v is increasing, so v is continuous at all but countably many points. It directly follows that f is continuous at all but countably many points.

For $N = 1$, we have $|f(x) - f(y)| \le v(y) - v(x)$, so $f(x) - f(y) \le v(y) - v(x)$ and $f(y) - f(x) \le v(y) - v(x)$, showing that $v \pm f$ are increasing. □

Corollary. Let $I \subset \mathbb{R}$ be an interval, then

1. $BV_{\text{loc}}(I)$ is the smallest vector space that contains all monotone functions $f : I \to \mathbb{R}$.
2. $BV(I)$ is the smallest vector space that contains all $f : I \to \mathbb{R}$ that is bounded and monotone.

Proof. 1. By previous theorems, $BV_{\text{loc}}(I)$ contains all monotone functions. Conversely, for any $f \in BV_{\text{loc}}(I)$, we have

$$f = v + (f - v),$$

where both functions are monotone.

2. Similar.

□

Corollary. If $f \in BV_{\text{loc}}(I)$, then f is differentiable a.e.

1.3 Absolutely continuous functions

Definition (absolutely continuous)**.** Let $I \subset \mathbb{R}$ be an interval. Say $f : I \to \mathbb{R}^N$ is **absolutely continuous** if for every $\varepsilon > 0$, there exists $\delta > 0$ such that for any finite collection of intervals $\{[a_i, b_i]\}_{i=1}^n$ with $[a_i, b_i] \subset I$ and disjoint interior, and $\sum_{i=1}^n (b_i - a_i) \le \delta$, we have $\sum_{i=1}^n \|f(b_i) - f(a_i)\| \le \varepsilon$.

Remark. Note that taking $n = 1$ in the previous definition gives uniform continuity. Therefore absolute continuity implies uniform continuity.

Example. Let $f : I \to \mathbb{R}^N$ be Lipshitz continuous with constant L, then $\sum_{i=1}^n \|f(b_i) - f(a_i)\| \leq L \sum_{i=1}^n (b_i - a_i)$. Therefore, Lipshitz continuity implies absolute continuity.

Theorem. Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}^N$ absolutely continuous. Suppose $f' = 0$ a.e. Then, f is constant.

Example. The Cantor function has zero derivative a.e. but it is not constant, so it is not absolutely continuous.

Proof. Let $a < b \in I$. We want to show that $f(a) = f(b)$. Let

$$G = \{x \in (a, b) : f'(x) = 0\}.$$

It follows that $\mathcal{L}^1(G) = b - a$. For any $x \in G$, we have $h^{-1}(f(x+h) - f(x)) \to 0$ as $h \to 0$. Given $\varepsilon > 0$, there exists $h_x > 0$ such that $\left\|h^{-1}(f(x+h) - f(x))\right\| \leq \varepsilon$ for all $0 < |h| \leq h_x$. Now let

$$\mathcal{F} = \{(x, x+h) : x \in G, 0 < h < h_x\}.$$

By a previous lemma, for any $\delta > 0$, there exists a finite and disjoint collection of intervals $\{(x_i, x_i + h_i)\}_{i=1}^n \subset \mathcal{F}$ with

$$\begin{aligned}
\mathcal{L}_o^1\left(G \cap \bigcup_{i=1}^n (x_i, x_i + h_i)\right) &\geq \mathcal{L}_o^1(G) - \delta, \\
\sum_{i=1}^n \mathcal{L}^1((x_i, x_i + h_i)) &\leq \mathcal{L}_o^1(G) + \delta.
\end{aligned}$$

Let $x_1 < \cdots < x_n$. It follows that

$$\begin{aligned}
\sum_{i=1}^n h_i &\geq \mathcal{L}_o^1\left(G \cap \bigcup_{i=1}^n (x_i, x_i + h_i)\right) \\
&\geq b - a - \delta.
\end{aligned}$$

Therefore, $(a, x_1) \cup (x_1 + h_1, x_2) \cup \cdots \cup (x_n + h_n, b)$ has measure at most δ. By absolutely continuity, we have

$$\|f(x_1) - f(a)\| + \|f(x_1 + h_1) + f(x_2)\| + \cdots + \|f(b) - f(x_n + h_n)\| \leq \varepsilon.$$

Moreover, $\|f(x_i + h_i) - f(x_i)\| \leq \varepsilon h_i$. It follows that

$$\begin{aligned}
\|f(b) - f(a)\| &\leq \varepsilon + \sum_{i=1}^n \varepsilon h_i \\
&\leq \varepsilon + \varepsilon(b - a) + \delta.
\end{aligned}$$

Letting $\delta \to 0$ and $\varepsilon \to 0$ shows $f(b) = f(a)$. □

Theorem. Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}^N$ be absoultely continuous. Then, $f \in BV_{\text{loc}}(I; \mathbb{R}^N)$. In particular, f is differentiable a.e.

Proof. Let $[a, b] \in I$. Let δ be in the definition of absolute continuity when applying it to f with $\varepsilon = 1$. Divide interval $[a, b]$ into segments of length δ. Call the segments $\{[x_{i-1}, x_i]\}_{i=1}^n$, where $n \leq \delta^{-1}(b - a) + 1 < \infty$. It follows that

$$\operatorname*{Var}_{[a,b]} f = \sum_{i=1}^{n} \operatorname*{Var}_{[x_{i-1},x_i]} f \leq n < \infty,$$

completing the proof. □

1.4 The fundamental theorem of calculus

Theorem (FTC for Lipshitz functions)**.** Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ be Lipshitz. Then,

$$f(b) - f(a) = \int_a^b f'(x)\, dx.$$

Proof. Note that f being Lipshitz implies $f \in BV_{\text{loc}}(I)$, so f is differentiable a.e. Define

$$g(x) = \begin{cases} f(x) & (a \leq x \leq b), \\ f(b) & (x \geq b). \end{cases}$$

Note that g is still Lipshitz and let L be the Lipshitz constant. We have

$$\lim_{n\to\infty} \frac{g(x + n^{-1}) - g(x)}{n^{-1}} = g'(x) \text{ a.e.}$$

Since g is continuous, g' is measurable. Moreover, g being Lipshitz implies $|g'| \leq L$. It then follows from Dominated Convergence Theorem that

$$\lim_{n\to\infty} \int_a^b \frac{g(x + n^{-1}) - g(x)}{n^{-1}}\, dx = \int_a^b g'(x)\, dx.$$

Through similar computation as in the Lebesgue differentiation theorem, for LHS we have

$$\begin{aligned} \lim_{n\to\infty} \int_a^b \frac{g(x + n^{-1}) - g(x)}{n^{-1}}\, dx &= \lim_{n\to\infty} f(b) - n \int_a^{a+n^{-1}} f(t)\, dt \\ &= f(b) - f(a), \end{aligned}$$

where the last equality is by continuity of f. □

Remark. Comparing this with the proof of Lebesgue differentiation theorem for monotone function, we know f is continuous and Lipshitz. Lipshitz continuity gives us bounded derivative and the ability to use Dominated Convergence Theorem, and continuity let us know the final term is $f(a)$.

Theorem (FTC 1)**.** Let $g : [a, b] \to \mathbb{R}$ integrable. Let $f(x) = \int_a^x g(t)\,dt$. Then f is absolutely continuous, and $f'(x) = g(x)$ for a.e. $x \in [a, b]$.

To show this we need a few lemmas.

Lemma. Let $E \subset \mathbb{R}^N$ be Lebesgue measurable, $g : E \to \mathbb{R}$ be integrable. Then for all $\varepsilon > 0$, there exists $\delta > 0$ such that $F \subset E$ with $\mathcal{L}^N(F) \leq \delta$ implies $\int_F |g|\,dx \leq \varepsilon$.

Lemma. Let $g : [a, b] \to \mathbb{R}$ integrable such that $\int_a^x g(t)\,dt = 0$ for all $x \in [a, b]$. Then, $g = 0$ a.e.

Proof. Exercise. □

Now we begin the proof for the first fundamental theorem of calculus.

Proof. First we show $f \in AC$. Fix $\varepsilon > 0$, and let δ be as in the first lemma. Suppose $\{(a_i, b_i)\}_{i=1}^n$ are disjoint sub-interval of $[a, b]$ with $\sum_{i=1}^n (b_i - a_i) \leq \delta$. Let $U = \bigcup_{i=1}^n (a_i, b_i)$. It follows that

$$\sum_{i=1}^n |f(b_i) - f(a_i)| = \sum_{i=1}^n \left| \int_{a_i}^{b_i} g(t)\,dt \right| \leq \sum_{i=1}^n \int_{a_i}^{b_i} |g(t)|\,dt = \int_U |g(t)|\,dt \leq \varepsilon,$$

as desired. Since $f \in AC$, it is differentiation a.e.

It remains to show that $f'(x) = g(x)$ a.e. First assume g is bounded with $|g| \leq M$. In this case, f is Lipshitz. Then for any $c \in [a, b]$, we have $f(c) - f(a) = \int_a^c f'(t)\,dt$ by FTC for Lipshitz functions. It follows that

$$\int_a^c g(t)\,dt = \int_a^c f'(t)\,dt.$$

By the second lemma above, we have $f' = g$ a.e.

For the next step assume g is non-negative. Define

$$g_n(t) = \begin{cases} g(t) & (g(t) \leq n), \\ 0 & (g(t) > n). \end{cases}$$

We then have $0 \leq g_n \leq n$. Moreover,

$$f(x) = \int_a^x g(t)\,dt = \int_a^x g_n(t)\,dt + \int_a^x (g(t) - g_n(t))\,dt.$$

Write $G_n(x) = \int_a^x g_n(t)\,dt$ and $H_n = \int_a^x (g(t) - g_n(t))\,dt$. By the previous step, we know that $G_n'(x) = g_n(x)$ a.e. We also know that $g \geq g_n$, so H_n is increasing. Then, H_n is differentiable a.e. and $f' = g_n + H_n' \geq g_n$ a.e. Letting $n \to \infty$ gives $f' \geq g$ a.e.

To get equality, we take integral and obtain $\int_a^b f'(x)\,dx \geq \int_a^b g(x)\,dx$. Since $g \geq 0$, f is also increasing. It follows that $\int_a^b f'(x)\,dx \leq f(b) - f(a)$. However $\int_a^b g(x)\,dx = f(b) - f(a)$. This shows that $\int_a^b (f' - g) = 0$. With $f' \geq g$, this implies $f' = g$ a.e.

To complete the proof for general g, consider $g = g^+ - g^-$. □

Theorem (FTC 2)**.** Let $f : [a, b] \to \mathbb{R}$, then $f \in AC$ iff the following holds:

1. f is differentiable a.e.
2. f' is Lebesgue integrable.
3. $f'(x) - f'(x_0) = \int_{x_0}^{x} f'(t)\,dt$ for all $x, x_0 \in [a, b]$.

Proof. Suppose the three given conditions holds. Let $F(x) = \int_a^x f'(t)\,dt$. By the previous theorem (FTC 1), we know $F \in AC$. Meanwhile, condition 3 implies $f(x) = f(a) + F(x)$. This shows that $f \in AC$.

Conversely, suppose now that $f \in AC$. We have shown before that f is differentiable a.e. Moreover, f is the difference of two increasing functions, while the derivative of increasing functions is integrable. This shows that f' is integrable.

Again let $F(x) = \int_a^x f'(t)\,dt$. By FTC 1, $F \in AC$ and $F' = f'$ a.e. Moreover $f - F \in AC$ and $(f - F)' = 0$ a.e. By previous theorem, this implies $f - F = c$ a.e. for some constant c. Therefore, $f(x) = \int_a^x f'(x)\,dt + c$ and the proof is complete. □

2 Change of variables

2.1 Mollifiers

Definition (support)**.** Let (X, d) a metric space, $E \subset X$, and $f : E \to \mathbb{R}$. The **support** of f is

$$\operatorname{supp} f = \overline{\{x \in E : f(x) \neq 0\}}.$$

Write the set of continuous function with compact support as $C_c(X)$. Also for open set $U \subset X$, write $C_c^k(U) = C_c(U) \cap C^k(U)$.

Exercise. Define

$$\varphi(x) = \begin{cases} e^{(\|x\|^2-1)^{-1}} & (\|x\| < 1), \\ 0 & (\|x\| \geq 1). \end{cases}$$

Show that $\varphi \in C_c^\infty(\mathbb{R}^N)$.

Definition (mollifier)**.** Let $\varphi \in C_c^\infty(\mathbb{R}^N)$ with $\operatorname{supp} \varphi \subset B[0,1]$ and $\int \varphi = 1$. For any $\varepsilon > 0$, define

$$\varphi_\varepsilon(x) = \varepsilon^{-N} \varphi(\varepsilon^{-1} x).$$

These functions are called **mollifiers**. Note that $\operatorname{supp} \varphi_\varepsilon \subset B[0, \varepsilon]$.

Exercise. Recall that for $\lambda > 0$, we have $\mathcal{L}^N(\lambda E) = \lambda^N \mathcal{L}^N(E)$. Show that $\int \varphi_\varepsilon = 1$.

Definition (mollification)**.** Let $E \subset \mathbb{R}^N$ be measurable, $f : E \to \mathbb{R}$ integrable. For $x \in \mathbb{R}^N$, define

$$f_\varepsilon(x) = \int_E \varphi_\varepsilon (x - y) f(y) \, dy.$$

The function f_ε is the **mollification** of f.

Theorem. Let $E \subset \mathbb{R}^N$ be measurable, $f : E \to \mathbb{R}$ integrable, then $f_\varepsilon \in C^\infty(\mathbb{R}^N)$, and

$$\frac{\partial^\alpha f_\varepsilon}{\partial x^\alpha}(x) = \int_E \frac{\partial^\alpha \varphi_\varepsilon}{\partial x^\alpha}(x - y) f(y) \, dy.$$

Proof. First show f_ε is continuous. Fix $x \in \mathbb{R}^N$ and let $\{x_n\}_{n=1}^\infty$ be a sequence with $x_n \to x$. Note that

$$|\varphi_\varepsilon(x_n - y) f(y)| \leq \sup |\varphi_\varepsilon| \, |f(y)| \, ,$$

which is integrable. It then follows from Dominated Convergence Theorem that $f_\varepsilon(x_n) = f(x)$ as $n \to \infty$.

For the paritial derivative, fix $i \in [N]$. We have

$$\begin{aligned} &\frac{f_\varepsilon(x + te_i) - f_\varepsilon(x)}{t} - \int_E \frac{\partial \varphi_\varepsilon}{\partial x_i}(x - y) f(y) \, dy \\ &= \int_E f(y) \left[\frac{\varphi_\varepsilon(x + te_i) - \varphi_\varepsilon(x - y)}{t} - \frac{\partial \varphi_\varepsilon}{\partial x_i}(x - y) \right] dy. \end{aligned}$$

By mean value theorem, we have

$$\varphi_\varepsilon(x+te_i-y)-\varphi_\varepsilon(x-y)=t\frac{\partial\varphi_\varepsilon}{\partial x_i}(x+\theta e_i-y),$$

for some $|\theta|\le|t|$. Hence the absolute value of the integrand is at most $2\,|f(y)|\sup|\partial_i\varphi_\varepsilon|$, which is integrable. Then Dominated Convergence Theorem applies and shows the formula for the paritial derivative.

By induction, we can continue to show that $f_\varepsilon\in C^\infty(\mathbb{R}^N)$ and $\partial_\alpha f_\varepsilon$ is as desired. □

Now we discuss some properties of the mollification.

Theorem. Let $E\subset\mathbb{R}^N$ be measurable and $f:E\to\mathbb{R}^N$ be integrable. Then the following holds:

1. Suppose f is continuous at $x\in E^\circ$. Then, $f_\varepsilon(x)\to f(x)$ as $\varepsilon\to0^+$.
2. Suppose f is continuous in E. Then for any compact $K\subset E^\circ$, $f_\varepsilon\to f$ uniformly in K.

Proof. 1. Since $x\in E^\circ$, there exists $B(x_0,r)\subset E$. For $\varepsilon<r$, we have

$$\begin{aligned}f_\varepsilon(x)-f(x)&=\int_{E\cap B(x,\varepsilon)}\varphi_\varepsilon(x-y)f(y)\,dy-f(x)\\&=\int_{B(x,\varepsilon)}\varphi_\varepsilon(x-y)f(y)\,dy-f(x)\\&=\int_{B(x,\varepsilon)}\varphi_\varepsilon(x-y)(f(y)-f(x))\,dy.\end{aligned}$$

Fix $\eta>0$. There exists $\delta>0$ such that $\|y-x\|<\delta$ implies $|f(y)-f(x)|<\eta$. Then when $\varepsilon<\delta$, we have

$$|f_\varepsilon(x)-f(x)|\le\eta\int_{B(x,\varepsilon)}\varphi_\varepsilon(x-y)\,dy=\eta.$$

The proof is then complete.

2. Let $K\subset E^\circ$ be compact. Let $\eta>0$ be given. Since f is continuous, there exists $\delta>0$ such that for any $x\in K$ and $y\in E$ with $\|x-y\|<\delta$, $|f(x)-f(y)|<\eta$. It follows that for $x\in K$ and $\varepsilon<\Delta$, we again have

$$\begin{aligned}f_\varepsilon(x)-f(x)&=\int_{E\cap B(x,\varepsilon)}f(y)\varphi_\varepsilon(x-y)\,dy-f(x)\\&=\int_{B(x,\varepsilon)}\varphi_\varepsilon(x-y)(f(y)-f(x))\,dy.\end{aligned}$$

Hence,

$$|f_\varepsilon(x)-f(x)|\le\eta\int_{B(x,\varepsilon)}\varphi_\varepsilon(x-y)\,dy=\eta.$$

Note that δ is not dependent on x. This implies that $f_\varepsilon \to f$ uniformly as $\varepsilon \to 0$.

□

Theorem (density)**.** Let $E \subset \mathbb{R}^N$ be measurable and $f : E \to \mathbb{R}^N$ be integrable. Then for every $\eta > 0$, there exists $g \in C_c^\infty(\mathbb{R}^N)$ such that

$$\int_E |f - g| \, dx < \eta.$$

In other words, $C_c^\infty(\mathbb{R}^N)$ is dense in $L^1(E)$.

Proof. For step 1, first suppose Q is a cube and $f = \chi_Q$. Then we have $f_\varepsilon \in C_c^\infty(\mathbb{R}^N)$. Also, $f_\varepsilon = 1$ if $x \in Q$ with $\operatorname{dist}(x, \partial Q) > \varepsilon$, and $f_\varepsilon = 0$ if $x \in Q^c$ with $\operatorname{dist}(x, \partial Q) > \varepsilon$. It follows that

$$\int_E |f_\varepsilon - f| \, dx \leq 2\mathcal{L}^N(\{x \in \mathbb{R}^N : \operatorname{dist}(x, \partial Q) \leq \varepsilon\}) \to 0$$

as $\varepsilon \to 0$, completing this step.

For step 2, suppose $\{Q_i\}_{i=1}^n$ are cubes with disjoint interior, $Q = \bigcup_{i=1}^m Q_i$, and $f = \chi_Q$. We can correspondingly write $f = \sum_{i=1}^n \chi_{T_i}$ where T_i are disjoint cubes. Applying step 1 gives the desired result, where $g = \sum_{i=1}^n (\chi_{T_i})_\varepsilon$.

For part 3, suppose U is open and $f = \chi_U$. Then we can write $U = \bigcup_{i=1}^\infty Q_i$, where $\{Q_i\}_{i=1}^\infty$ are cubes with disjoint interior. Write $F_n = \bigcup_{i=1}^n Q_i$. Note that f is integrable, so Dominated Convergence Theorem implies

$$\lim_{n\to\infty} \int_E |f - \chi_{F_n}| \, dx = 0.$$

Let n be so large that $\int_E |f - \chi_{F_n}| \, dx < \eta$. Applying step 2 to find $g \in C_c^\infty(\mathbb{R}^N)$ such that $\int_E |g - \chi_{F_n}| \, dx < \eta$. This gives the desired result. Note that now g is not the mollification of f.

For step 4, suppose G is measurable and $f = \chi_G$, then there exists $U \supset G$ open with $\mathcal{L}^N(U \backslash G) < \eta$. Step 3 gives $g \in C_c^\infty(\mathbb{R}^N)$ such that $\int_E |g - \chi_U| \, dx < \eta$. Moreover, $\int_E |\chi_U - \chi_G| = \mathcal{L}^N(U \setminus G) < \eta$, so this g is as desired.

For step 5, we can show the result for simple functions. Then we can prove the result for non-negative functions by approximating it with simple functions and using Dominated Convergence Theorem. Finally, for general function f, writing $f = f^+ - f^-$ and using previous steps complete the proof. □

Theorem. Let $E \subset \mathbb{R}^N$ be measurable and $f : E \to \mathbb{R}$ integrable. Then there exists a sequence of functions $\{f_n\}_{n=1}^\infty \subset C_c^\infty(\mathbb{R}^N)$ such that $f_n \to f$ pointwise a.e. and in L^1. That is,

$$\lim_{n\to\infty} \int_E |f_n - f| \, dx = 0.$$

Lemma. Let $E \subset \mathbb{R}^N$ be measurable and $f_n, f : E \to \mathbb{R}$ integrable with

$$\lim_{n\to\infty} \int_E |f_n - f| \, dx = 0.$$

Then there exists a subsequence $\{f_{n_k}\}_{k=1}^{\infty}$ such that $f_{n_k} \to f$ pointwise a.e.

Proof. Given $k \in \mathbb{N}$, we can find $n_k \in \mathbb{N}$ such that

$$\int_E |f_n - f| \, dx \le 2^{-k}$$

for all $n \ge n_k$. WLOG assume $n_k < n_{k+1}$. Define $g = \sum_{k=1}^{\infty} |f_{n_k} - f|$. It follows that

$$\int_E g = \sum_{k=1}^{\infty} \int_E |f_{n_k} - f| \le \sum_{k=1}^{\infty} 2^{-k} < \infty.$$

Hence, $g < \infty$ a.e. and thus $|f_{n_k} - f| \to 0$ a.e. □

Proof. Use the previous theorem to find a sequence that converges in L^1, then apply the lemma to find a subsequence that converges pointwise a.e. □

Remark. Suppose f is bounded with $|f(x)| \le M$. Can we make f_n also bounded? First consider relaxing the smooth requirement to just continuous. We can define $h_n = (f_n \wedge M) \vee (-M)$. Then h_n are continuous, $h_n \to f$ pointwise, and $|h_n(x)| \le M$. Using a similar idea, we can make h_n smooth. Therefore, when f is bounded we can assume the converging sequence is also uniformly bounded with the same constant.

Suppose $E = K$ is compact. Can we make the $\operatorname{supp} f_n \subset K_n$ where K_n are compact and shrink to E? Consider the mollification $\chi_{K_\delta} * \varphi_\varepsilon$, and multiply it with f_n. Verify that $\varepsilon = \delta = n^{-1}$ works as desired.

2.2 Change of variables

Lemma (CoV for linear transformation)**.** Let $L : \mathbb{R}^N \to \mathbb{R}^N$ be linear and invertible. Write $L(x) = Ax$ where A is a matrix. Suppose $f : \mathbb{R}^N \to \mathbb{R}$ integrable. Then,

$$\int f(y) \, dy = |\det A| \int f(L(x)) \, dx.$$

Proof. The linear map L is given by the composition of a finite number of transformation of the type $t_s(x) = (sx_1, x_2, \ldots, x_N)$ where $s \neq 0$, $a(x) = (x_1 + x_2, x_2, \ldots, x_N)$, and $s_{ij}(x)$ which swaps the i and j components of x. Hence, it suffices to show the lemma where L is each of these operation.

Assume first $f \geq 0$. For the first operation t_s, we can verify that for integrable $g : \mathbb{R} \to \mathbb{R}$ and $s \neq 0$, we have $\int g(y)\,dy = |s| \int g(sx)\,dx$ (Exercise). It follows from Tonelli's theorem that

$$\begin{aligned}
|\det t_s| \int_{\mathbb{R}^N} f(t_s(x))\,dx &= |s| \int_{\mathbb{R}^N} f(sx_1, x_2, \ldots, x_n)\,dx \\
&= |s| \int_{\mathbb{R}} \int_{\mathbb{R}^{N-1}} f(sx_1, x_2, \ldots, x_n)\,dx_1\,dx_{\setminus 1} \\
&= \int_{\mathbb{R}} \int_{\mathbb{R}^{N-1}} f(y_1, x_2, \ldots, x_n)\,dy_1\,dx_{\setminus 1} \\
&= \int_{\mathbb{R}^N} f(y)\,dy.
\end{aligned}$$

For the second operation a, we have $\int g(y)\,dy = \int g(x+s)\,dx$ for $s \in \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ integrable (Exercise). Then Tonelli's theorem gives

$$\begin{aligned}
\int_{\mathbb{R}^N} f(a(x))\,dx &= \int_{\mathbb{R}^{N-1}} \int_{\mathbb{R}} f(x_1 + x_2, x_2, \ldots, x_N)\,dx_1\,dx_{\setminus 1} \\
&= \int_{\mathbb{R}^{N-1}} \int_{\mathbb{R}} f(x_1, x_2 \ldots, x_N)\,dx_1\,dx_{\setminus 1} \\
&= \int_{\mathbb{R}^N} f(y)\,dy.
\end{aligned}$$

The claim for the third operation s_{ij} follows directly from Tonelli's theorem. □

Theorem (CoV for continuous functions)**.** Let $U \subset \mathbb{R}^N$ be open and $g : U \to \mathbb{R}^N$ be injective and C^1. Suppose that $\det Dg(x) \neq 0$. Let $f : g(U) \to \mathbb{R}$ be continuous and with compact support. Then

$$\int_{g(U)} f(y)\,dy = \int_U f(g(x))\,|\det Dg(x)|\,dx.$$

Remark. Note that by inverse function theorem, $g(U)$ is open.

Lemma. Let $U \subset \mathbb{R}^N$ be open, $g : U \to \mathbb{R}^N$ be injective and C^1. Suppose that $\det Dg(x) \neq 0$ for all $x \in U$. For any $K \subset g(U)$ compact, we have

$$\lim_{\varepsilon \to 0^+} \int_U \varphi_\varepsilon(g(x) - y)\,dx = \frac{1}{|\det Dg(g^{-1}(y))|}$$

for all $y \in K$.

Proof. For step 1, recall that $\operatorname{supp} \varphi_\varepsilon \subset B[0, \varepsilon]$, so $\varphi_\varepsilon(g(x) - y) = 0$ if $\|g(x) - y\| > \varepsilon$. Define

$$K_\varepsilon = \left\{ y \in \mathbb{R}^N : \operatorname{dist}(y, K) \leq \varepsilon \right\},$$

which is compact. Since $g(U)$ is open and compact $K \subset g(U)$, we have $K_\varepsilon \subset g(U)$ for ε small enough. By inverse function theorem and the fact that g is injective, there exists $g^{-1} : g(U) \to \mathbb{R}^N$ that is C^1. Hence there exists some constant $L \geq 1$ so that $\left\|Dg^{-1}(y)\right\| \leq L$ for all $y \in K_\varepsilon$. Now fix $z \in K$. For $y \in \mathbb{R}^N$ with $\|y - z\| \leq \varepsilon$, we have $y \in K_\varepsilon$ and by mean value theorem,

$$\left\|g^{-1}(y) - g^{-1}(z)\right\| \leq L \|y - z\| .$$

Now consider any $x \in U$. If $\|g(x) - z\| \leq \varepsilon$, $\left\|x - g^{-1}(z)\right\| \leq L\varepsilon$. Otherwise, if $\|g(x) - z\| > \varepsilon$, $\varphi_\varepsilon(g(x) - z)\, dx = 0$. Write. Hence,

$$\begin{aligned}\int_U \varphi_\varepsilon(g(x) - z)\, dx &= \int_{B[g^{-1}(z), L\varepsilon]} \varphi_\varepsilon(g(x) - z)\, dx \\ &= \frac{1}{\varepsilon^N} \int_{B[g^{-1}(z), L\varepsilon]} \varphi\left(\frac{g(x) - z}{\varepsilon}\right) dx.\end{aligned}$$

Use linear change of variables with $x = g^{-1}(z) + \varepsilon u$ and thus $dx = \varepsilon^N du$. We obtain

$$\int_U \varphi_\varepsilon(g(x) - z)\, dx = \int_{B[0,L]} \varphi\left(\frac{g(g^{-1}(z) + \varepsilon u) - g(g^{-1}(z))}{\varepsilon}\right) du.$$

Note that φ is bounded and $B[0, L]$ is also bounded, Dominated Convergence Theorem applies and we can take $\varepsilon \to 0^+$. Note also $L \geq 1$ while $\operatorname{supp} \varphi \subset B[0, 1]$, so

$$\begin{aligned}\int_U \varphi_\varepsilon(g(x) - z)\, dx &= \int_{B[0,L]} \varphi\left(Dg(g^{-1}(z))u\right) du \\ &= \int_{\mathbb{R}^N} \varphi\left(Dg(g^{-1}(z))u\right) du.\end{aligned}$$

Using linear change of variable again, we obtain

$$\int_U \varphi_\varepsilon(g(x) - z)\, dx = \frac{1}{|\det Dg(g^{-1}(z))|} \int_{\mathbb{R}^N} \varphi(x)\, dx = \frac{1}{|\det Dg(g^{-1}(z))|}.$$

□

Theorem (CoV for continuous functions)**.** Let $U \subset \mathbb{R}^N$ be open and $g : U \to \mathbb{R}^N$ be injective and C^1. Suppose that $\det Dg(x) \neq 0$. Let $f : g(U) \to \mathbb{R}$ be continuous and with compact support. Then

$$\int_{g(U)} f(y)\, dy = \int_U f(g(x)) \left|\det Dg(x)\right| dx.$$

Proof sketch. Write $K = \operatorname{supp} f \subset g(U)$. Choose $\varepsilon > 0$ small enough so that $K_\varepsilon \subset g(U)$, where as usual $K_\varepsilon = \left\{y : \mathbb{R}^N : \operatorname{dist}(y, K) \leq \varepsilon\right\}$. For $y \in g(U)$, define

$$p(y) = f(y) \left|\det Dg(g^{-1}(y))\right| .$$

We have $\operatorname{supp} p \subset K$, so $\operatorname{supp} p_\varepsilon \subset K_\varepsilon$. Note that p is continuous and thus $p_\varepsilon \to p$ uniformly on compact sets, i.e. $p_\varepsilon \to p$ uniformly on K_ε. It is also easy to verify that $p_\varepsilon \circ g \to p \circ g$ pointwise. It then follows from Dominated Convergence Theorem that

$$\int_U p_\varepsilon(g(x))\,dx = \int_{g^{-1}(K_\varepsilon)} p_\varepsilon(g(x))\,dx \to \int_U f(g(x))\,|\det Dg(x)|\,dx$$

as $\varepsilon \to 0^+$. On the other hand, Fubini's Theorem gives

$$\begin{aligned}\int_U p_\varepsilon(g(x))\,dx &= \int_U \int \varphi_\varepsilon(g(x)-y)p(y)\,dy\,dx \\ &= \int \int_U \varphi_\varepsilon(g(x)-y)p(y)\,dx\,dy\end{aligned}$$

Using the lemma, we have

$$\int_U p_\varepsilon(g(x))\,dx = \int \frac{1}{|\det Dg(g^{-1}(y))|} f(y)\left|\det Dg(g^{-1}(y))\right|\,dy = \int_{g(U)} f(y)\,dy.$$

□

Exercise. Fill in the details of the proof above.

Next we want to prove a stronger version, whre we do not require f to be continuous and with compact support, but only need f to be integrable.

Theorem (CoV)**.** Let $U \subset \mathbb{R}^N$ be open and $g : U \to \mathbb{R}^N$ be injective and C^1. Suppose that $\det Dg(x) \neq 0$. Suppose $g(U) \subset E$ where E Lebesgue measurable, $f : E \to \mathbb{R}$ integrable. Then

$$\int_{g(U)} f(y)\,dy = \int_U f(g(x))\,|\det Dg(x)|\,dx.$$

Proof. For step 1, assume $E = K$ compact and $|f(x)| \leq M$. Find $f_n \in C_c^\infty(\mathbb{R}^N)$ such that $f_n \to f$ pointwise a.e. Moreover, referring to remark on a previous theorem, assume $|f_n(x)| \leq 2M$ and $\operatorname{supp} f_n \subset K_{n^{-1}} \subset g(U)$. By change of variable,

$$\int_{g(U)} f_n(y)\,dy = \int_U f_n(g(x))\,|\det Dg(x)|\,dx.$$

Note that $K_{n^{-1}} \subset K_\delta \subset g(U)$. Therefore $\int_{K_\delta} f_n(y)\,dy = \int_{g^{-1}(K_\delta)} f_n(g(x))\,|\det Dg(x)|\,dx$. Applying Dominated Convergence Theorem, we obatin

$$\int_K f(y)\,dy = \int_{g^{-1}(K)} f_n(g(x))\,|\det Dg(x)|\,dx.$$

For step 2, still assume $f \geq 0$ bounded. We can write $E = \bigcup_{n=1}^{\infty} K_n \cup E_0$ where K_n increasing and compact, and $\mathcal{L}^N(E_0) = 0$. Then we have $\mathcal{L}^N(g^{-1}(E_0)) = 0$, and

$$\int_{E_0} f \, dy = \int_{g^{-1}(E_0)} f(g(x)) \left|\det Dg(x)\right| \, dx = 0.$$

In K_n we use step 1, and because $f \geq 0$, apply Monotone Convergence Theorem.

Finally, to remove bounded requirement, consider $f_n = f \wedge n$ and use Monotone Convergence Theorem. For general real-valued function, write $f = f^+ + f^-$ and apply previous steps. $\square$